Homework 2

ORF 245 - Fundamentals of Statistics

Due: Feb. 14th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
- Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).

Exercise: 1. (Counting Techniques for Passwords) Suppose we restrict a password to use only lower case letters of the English alphabet or one of the 10 digits (so $26 + 10 = 36$ possibilities for each character of the password).

Note: You do not need to simplify expressions and compute exact values. You may leave your answer in terms of permutations, combinations, factorials, or products.

(a) How many passwords of size 6 can be generated?

- **Solution**:
  Consider 6 sub-experiments where each consists of choosing a character. We have 36 choices for each character, and therefore, the result is 36^6 possible outcomes for a 6-character password.

(b) How many passwords of size 8, such that no characters are repeated, can be generated?

- **Solution**:
  The order of characters in a password matters, and here we assume that characters are sampled without replacement. Therefore, the result is $P(36, 8)$.

(c) Suppose that the first symbol of the password is restricted to be a. How many passwords of (total) size 6, such that no letters are repeated, can be generated from lower case letters of English alphabet?

- **Solution**:
  The remaining 5 characters must come from the 25 remaining letters. Since order again matters, the result is $P(25, 5)$.

(d) Suppose that we require one of the symbols in the password to be a, but it can be located anywhere in the password. How many such passwords of size 6, in which no characters are repeated, are possible?

- **Solution**:

  The result is $6 \times P(35, 5)$ – since there are 6 possible locations for the character a to occur, and for each such location we can obtain $P(35, 5)$ unique passwords by choosing 5 unique characters from the remaining 35 (where the order of these characters matters).

Exercise: 2. (Conditional Probability) Suppose we are given the following events and probabilities

Events	A	B	C	$A \cap B$	$A \cap C$	$B \cap C$	$A \cap B \cap C$
Probabilities	0.14	0.23	0.37	0.08	0.09	0.13	0.05

(a) Compute $\mathbb{P}(C|B)$ and $\mathbb{P}(B|C)$.

- **Solution**:
  $\mathbb{P}(C|B) = \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(B)} = \frac{0.13}{0.23} \approx 0.57.$
  $\mathbb{P}(B|C) = \frac{\mathbb{P}(B \cap C)}{\mathbb{P}(C)} = \frac{0.13}{0.37} \approx 0.35.$

(b) Compute $\mathbb{P}(A \cap C|B \cap A)$.

- **Solution**:
  $\mathbb{P}(A \cap C|B \cap A) = \frac{\mathbb{P}((A \cap C) \cap (B \cap A))}{\mathbb{P}(B \cap A)} = \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(A \cap B)} = \frac{0.05}{0.08} = 0.625.$

(c) Compute $\mathbb{P}(B|B)$.

- **Solution**:
  $\mathbb{P}(B|B) = 1$, since $B \cap B = B$ and using the definition of conditional probability.

(d) Compute $\mathbb{P}(B|A \cup B)$.

- **Solution**:
  Using the Inclusion-Exclusion Principle:
  $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = 0.14 + 0.23 - 0.08 = 0.29.$
  $\mathbb{P}(B|A \cup B) = \frac{\mathbb{P}(B \cap (A \cup B))}{\mathbb{P}(A \cup B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(A \cup B)} = \frac{0.23}{0.29} \approx 0.79.$

Exercise: 3. (Losing Your Marbles) Suppose there are two bags with three marbles each. One of the bags (call it Bag 0) has three red marbles and no white marbles, and the other bag (call it Bag 1) has two red marbles and one white marble.

One of the two bags is chosen at random, where each bag is equally likely. You don't know which bag has been chosen, but you do get to pull out two marbles (one at a time) from the bag *without replacement* and observe their colors.

(a) Conditional on Bag 0 being chosen, what is the probability that the first marble pulled out is red?

- **Solution**:
  We denote B as the number of the bag being picked($B = 0$ represents Bag 0). We also denote R as the event that the first marble pulled out is red and $2R$ as the event that both two marbles pulled out are red.
  $\mathbb{P}(R|B = 0) = \frac{3}{3} = 1$ since all three marbles in Bag 0 are red.

(b) What is the probability that the first marble you pull out is red? (Without conditioning on Bag 0 being chosen).

- **Solution**:
  Using the law of total probability:
  $\mathbb{P}(R) = 0.5 \times \mathbb{P}(R|B = 0) + 0.5 \times \mathbb{P}(R|B = 1) = 0.5 + 0.5 \times \frac{2}{3} = \frac{5}{6}$.

(c) What is the probability that both the first and the second marble you pull out are both red?

- **Solution**:
  Again, using the law of total probability:
  $\mathbb{P}(2R) = 0.5 \times \mathbb{P}(2R|B = 0) + 0.5 \times \mathbb{P}(2R|B = 1) = 0.5 + 0.5 \times \frac{2}{3} \times \frac{1}{2} = \frac{2}{3}$

(d) What is the probability that the bag chosen (i.e., from which you pulled out the marbles) is Bag 0 given that the two marbles you pulled out are both red?

- **Solution**:
  By using Bayes' Rule:
  $\mathbb{P}(B = 0|2R) = \frac{\mathbb{P}(2R|B=0)\mathbb{P}(B=0)}{\mathbb{P}(2R)} = \frac{0.5}{2/3} = \frac{3}{4} = 0.75$

Exercise: 4. (Expected Value) Recall that for a discrete random variable X with probability mass function (pmf) given by $p(x)$, the expected value of X (denoted $\mathbb{E}[X]$) and the variance of X (denoted $\mathrm{Var}(X)$) are defined as:

$$\mathbb{E}[X] = \sum_{x \in \mathrm{Range}(X)} x \cdot p(x) \qquad \mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

Consider a random variable X with the following pmf:

x	-3	0	3	8
$p(x)$	$1/4$	$1/2$	$1/8$	$1/8$

(a) Compute $\mathbb{E}[X]$.

- **Solution**:
  $\mathbb{E}[X] = \frac{-3}{4} + \frac{0}{2} + \frac{3}{8} + \frac{8}{8} = \frac{5}{8}$.

(b) Find the pmf of another random variable $Y = X^2$. Compute its expected value $\mathbb{E}[Y]$.

- **Solution**:
  Let $Y = X^2$. The pmf of Y is as follows:

y	0	9	64
$p_Y(y)$	$1/2$	$1/4 + 1/8 = 3/8$	$1/8$

where $p_Y(y) = 0$ for all other values.
Therefore, $\mathbb{E}[Y] = \frac{9}{4} + \frac{0}{2} + \frac{9}{8} + \frac{64}{8} = \frac{91}{8}$.

(c) Compute $\mathrm{Var}(X)$.

- **Solution**:
  $\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[Y] - \mathbb{E}[X]^2 = \frac{86}{8} = 10.75.$

(d) Compute $\mathbb{E}[8X + 1]$.

- **Solution**:
  By linearity of expectation, we can compute:
  $\mathbb{E}[8X + 1] = 8\mathbb{E}[X] + 1 = 6$

Exercise: 5. [OPTIONAL] (Discrete Random Variables)
Note: This is an ***optional*** exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Consider tossing a fair coin (equal likelihood of Heads / Tails), and let X be the toss on which the first head appears. Let $Y = \min\{X, 4\}$. That is, $Y = X$ if $X = 1, 2, 3$, and $Y = 4$ otherwise.

(a) Find the probability mass function (pmf) of X.

- **Solution**:
  Since X represents the first success in a sequence of independent Bernoulli trials with probability $p = \frac{1}{2}$, it follows a geometric distribution:

$$P(X = k) = (1-p)^{k-1}p = \left(\frac{1}{2}\right)^{k-1}\frac{1}{2} = \left(\frac{1}{2}\right)^{k}, \quad k = 1, 2, 3, \ldots$$

(b) Find the pmf of Y.

- **Solution**:
  Since Y takes values $1, 2, 3$ directly from X and groups all larger values into 4, we compute:

$$P(Y = k) = P(X = k) = \left(\frac{1}{2}\right)^{k}, \quad k = 1, 2, 3$$

  For $Y = 4$, it accounts for all cases where $X \geq 4$:

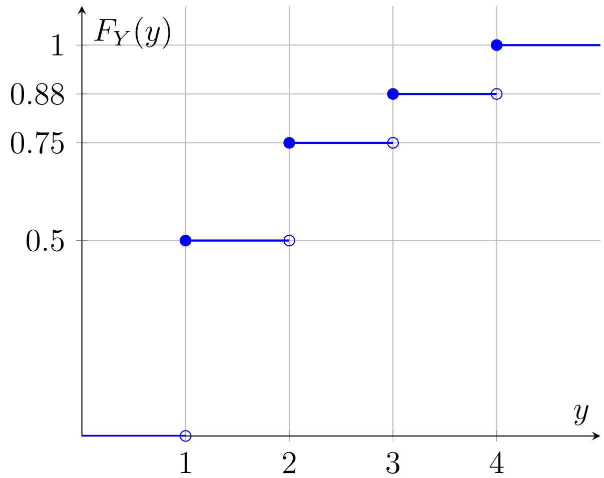
$$\begin{aligned} P(Y = 4) = P(X \geq 4) &= 1 - P((X = 1) \cup (X = 2) \cup (X = 3)) \\ &= 1 - P(X = 1) - P(X = 2) - P(X = 3) \end{aligned}$$

$$= 1 - \frac{1}{2} - \frac{1}{4} - \frac{1}{8} = \frac{1}{8}.$$

(c) Sketch the cumulative distribution function (cdf) of Y.

- **Solution**:
  The cumulative distribution function $F_Y(y)$ is:

$$F_Y(y) = \begin{cases} 0, & y < 1 \\ \frac{1}{2}, & 1 \leq y < 2 \\ \frac{3}{4}, & 2 \leq y < 3 \\ \frac{7}{8}, & 3 \leq y < 4 \\ 1, & y \geq 4. \end{cases}$$



(d) Let A be the event that Y is odd, B be the event that Y is even. Are A and B independent events? Justify your answer.

- **Solution**:
  Define events:

$$A = \{Y \text{ is odd}\} = \{Y = 1, 3\}$$
$$B = \{Y \text{ is even}\} = \{Y = 2, 4\}$$

  Compute probabilities:

$$P(A) = P(Y = 1) + P(Y = 3) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$
$$P(B) = P(Y = 2) + P(Y = 4) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

  Since A and B are complementary ($P(A \cup B) = 1$, and $P(A \cap B) = 0$), they are not independent.

(e) Find $\mathbb{E}[Y]$.

- **Solution**:

$$E[Y] = 1 \cdot P(Y = 1) + 2 \cdot P(Y = 2) + 3 \cdot P(Y = 3) + 4 \cdot P(Y = 4)$$
$$= 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} + 4 \cdot \frac{1}{8}$$
$$= \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{8}$$
$$= \frac{4}{8} + \frac{4}{8} + \frac{3}{8} + \frac{4}{8} = \frac{15}{8}.$$

  Thus, $E[Y] = \frac{15}{8} = 1.875$.